

Revisiting Metaheuristic Comparisons for Continuous Wind Farm Layout Optimization: Baseline Configuration, Component Analysis and a Two-Phase PSO–VNS Design

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Abstract—Wind farm layouts fix a plant’s lifetime energy yield, yet comparisons of layout optimizers are hard to trust. This paper shows that, in continuous wind farm layout optimization, their conclusions depend strongly on baseline configuration and comparison control. Under a controlled protocol (68 benchmark cases, 6,030 evaluations including gradient evaluations, seed-paired runs, feasibility-aware ranking, Holm-corrected tests), a convergent particle swarm optimization (PSO) followed by basic variable neighborhood search (PSO-VNS) is the most consistent of eight methods, while salp-swarm phases add little beyond random sampling and the Laplace step makes the hybrid worse. First, the PSO setting $w = 0.7$, $c_1 = c_2 = 2$ violates the order-2 stability condition; with it, PSO is feasible in 86.0% of the runs and ranks behind salp-swarm hybrids, whereas with the constriction setting it ranks first. Second, against a random-sampling control, a PSO phase gives VNS a clearly better start than salp-swarm phases, and the VNS phase significantly improves only the salp-swarm methods. Third, PSO-VNS has the best average rank (1.82), is feasible in every run, and has the highest mean yield on the Horns Rev 1 block at 6,030 and 30,030 evaluations; however, it is not significantly different from PSO overall ($p = 0.30$), VNS ranks first from feasible starts and PSO under a Gaussian re-evaluation of the final layouts, and a gap to the best published IEA Wind Task 37 layouts remains.

Index Terms—Benchmarking, metaheuristics, particle swarm optimization, variable neighborhood search, wake effect, wind farm layout optimization.

NOMENCLATURE

B	Budget (number of evaluations).
C_T, K	Thrust coefficient; Jensen wake-spreading constant.
D, R	Rotor diameter and rotor radius ($D = 2R$).

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E_{ideal}	Wake-free benchmark objective of one turbine.
F_p	Penalized wake loss (minimized).
$\ell_{\min}$	Minimum spacing between turbines.
N, N_p	Number of turbines; population size.
N_θ, N_s	Numbers of wind-direction and wind-speed bins.
r	Farm radius.
w, c_1, c_2	Inertia weight and acceleration coefficients of PSO.
$\mathbf{X}$	Layout vector $(x_1, y_1, \dots, x_N, y_N)$.
θ	Wind direction (blowing towards, CCW from east).
ω	Fraction of the budget given to Phase 1.

I. INTRODUCTION

The positions of the turbines in a wind farm are fixed at the planning stage and determine, for the lifetime of the farm, how much of the available wind energy it converts. An upstream turbine leaves behind a wake of slower and more turbulent air that reduces the power of the turbines downstream, so the layout sets the wake loss and thereby the gross energy yield, and it also shapes the collector system and grid connection [1], [2]. As the wake loss is a share of the wake-free energy, a difference of a few tenths of a percentage point between two optimizers is a lifetime energy difference of that size. The wind farm layout optimization problem (WFLOP) places a prescribed number of turbines in a site to minimize the wake loss subject to boundary and minimum-spacing constraints; we address only this wake-loss part of layout design, for farms of up to 36 turbines. Its nonlinear objective, the coupling of all turbines by the wakes and the nonconvex spacing constraints have made the WFLOP a popular test bed for metaheuristics.

Early studies placed turbines on a grid with genetic algorithms (GAs) [3], [4]. The continuous, wind-distribution-based model of Kusiak and Song [5], which we use because it is a common reference for metaheuristics, has been solved with ant colony optimization and particle filtering [6], [7] and biogeography-based optimization [8]; GA, particle swarm optimization (PSO), differential evolution (DE) and simulated annealing were compared in [9], and PSO has been used to reduce wake losses [10]. Later work used informed GAs [11], [12], deterministic refinement [13], variable neighborhood search (VNS) for large offshore farms [14],

pseudo-gradients [15], Gaussian wake models [16] and surrogates [17]–[19]. The IEA Wind Task 37 case study [20] showed that, even with a common wake model, different optimizers reach clearly different layouts.

Many of these studies introduce a new or modified meta-heuristic and compare it with reference methods; the benchmarking literature notes that the outcome of such comparisons depends on how they are set up [21], [22]. Four issues matter for the WFLOP. First, a baseline may be configured outside its working range: the expected positions of PSO particles converge (order-1 stability) for a wide range of the inertia weight w and the acceleration coefficients c_1 , c_2 , but their variance converges (order-2 stability) only in a smaller region [23], [24], and the setting $w = 0.7$, $c_1 = c_2 = 2$, which keeps the acceleration coefficients of the original PSO [25], lies outside it. Second, budgets counted in iterations, or without the gradient evaluations of gradient-based solvers, are not equal. Third, penalty-based metaheuristics may return infeasible layouts, which must be ranked consistently. Fourth, conclusions drawn from mean values without paired tests and multiplicity correction [26], [27] may not hold. For two-phase hybrids, a further question is whether the first phase contributes more than a random start would.

This paper revisits such comparisons for the continuous WFLOP under a protocol that controls these factors. Our thesis is that their conclusions depend strongly on how the baselines are configured and how the comparison is controlled. Under a controlled protocol, a simple two-phase design, a convergent PSO followed by the basic VNS [28], [29] (PSO-VNS), is the most consistent of the eight methods compared, while salp-swarm phases add little beyond random sampling and the Laplace step makes the hybrid worse. The salp-swarm methods are the salp swarm algorithm (SSA) [30], [31] and the Laplacian SSA (LX-SSA) [32], proposed earlier by two of the authors; PSO-VNS is a sequential hybrid in the memetic sense, not a new PSO variant. Our contributions are as follows:

- 1) The application of established benchmarking practice [22], [26] to the continuous WFLOP (equal budgets that count every objective evaluation, including gradient evaluations, seed-paired runs, case-mean and run-level tests with Holm correction, re-evaluation under alternative modeling assumptions, public code and data), with two WFLOP-specific elements: feasibility-aware ranking and a random-sampling control for two-phase hybrids. We apply it to eight methods on 68 benchmark cases, a 16-turbine block of the Horns Rev 1 offshore farm and IEA Wind Task 37 (IEA37) Case Study 1 (Sections V, VIII and IX).
- 2) Evidence that the PSO baseline setting changes the order of PSO and the salp-swarm methods: this setting violates the order-2 stability condition; with it, PSO is feasible in 86.0% of the runs and, in a comparison of salp-swarm hybrids with standard baselines, ranks behind both hybrids; with the constriction setting [33] it ranks first there and second in the main comparison, ahead of every salp-swarm method, at 6,030 evaluations. At 120,030 evaluations, SSA-VNS ranks ahead of PSO on the six largest cases (average rank 4.17 vs 5.83;

Sections III-A, VI-A and VIII-A, Table I).

- 3) A component analysis of two-phase swarm-VNS hybrids with a random-sampling control (RS-VNS), a minimum bar, as uniform samples in the bounding square are rarely feasible. Against RS-VNS, an SSA phase gives only a small gain (tied at run level, 2/66/0; case-mean difference -0.068 pp, $p = 4.5 \times 10^{-4}$), an LX-SSA phase none ($p = 0.72$), and the Laplace step makes the hybrid worse (SSA-VNS better than LX-SSA-VNS, $p = 5.4 \times 10^{-6}$); a PSO phase gives a clearly larger gain. The VNS phase significantly improves both salp-swarm methods (39/29/0 and 35/33/0), but PSO only slightly (item 4; Section VII).
- 4) PSO-VNS as a simple reference method, with an account of where it wins and where it does not. It has the best average rank (1.82) and is feasible in every run, but over the 68 cases it is not significantly different from PSO ($p = 0.30$; significantly better in 14 cases and worse in 4 at run level), and in an exploratory split its advantage is concentrated in the larger layouts ($N \geq 10$) of Data Set II. On the six largest cases, it has the best average rank at all three budgets (exploratory, no test; first in 3 of the six cases at 120,030 evaluations), and on the Horns Rev 1 block it has the highest mean yield at 6,030 and 30,030 evaluations, with 3 of 30 runs at 30,030 evaluations above the modeled yield of the installed block (installed spacing $7D$, optimized $4D$). Sections VIII and IX show where it is not first: from feasible starts (VNS), under a Gaussian re-evaluation of the final layouts (PSO) and against the best published IEA37 layouts. From these results we derive recommendations for benchmarking WFLOP methods (Section X).

Sections II–IV describe the model and the methods, Section V the evaluation protocol, Sections VI–IX the results, Section X the recommendations, Section XI the limitations and Section XII the conclusions.

II. PROBLEM MODELING

For a prescribed number N of identical turbines, the $n = 2N$ coordinates (x_i, y_i) are optimized to maximize the expected farm power, i.e., to minimize the wake loss, under the Jensen wake model [34], [35] and the power curve of [5]. The wind direction θ has the probability density p_θ , and for each direction the wind speed s follows a Weibull density $p_s(s; \zeta, \psi)$ with shape $\zeta(\theta)$ and scale $\psi(\theta)$. Here θ is the direction *towards* which the wind blows, counter-clockwise from east; the meteorological directions θ_{met} of Horns Rev 1 and IEA37 (wind *from*, clockwise from north) are converted by $\theta = 270^\circ - \theta_{\text{met}}$. Two constraints apply: every pair of turbines satisfies $\ell_{ij} = \|(x_i - x_j, y_i - y_j)\|_2 \geq \ell_{\min} = 4D = 8R$, as in [5], [8], and the farm is circular, $x_i^2 + y_i^2 \leq r^2$, with $r = 500, 750$ or 1000 m.

A. Wake Effect Model

In the Jensen model (Figs. S13–S15 of the supplementary material), the wake of turbine j is a half cone with half-angle $\alpha = \arctan K$ (wake-spreading constant K), and its virtual

vertex lies R/K upstream of the rotor. With $u_{ij} = (x_i - x_j) \cos \theta + (y_i - y_j) \sin \theta + R/K$ and the lateral offset $v_{ij} = -(x_i - x_j) \sin \theta + (y_i - y_j) \cos \theta$, turbine i lies in the wake if $\beta_{ij} = \cos^{-1}(u_{ij}/\sqrt{u_{ij}^2 + v_{ij}^2}) < \alpha$. Its downstream distance is $d_{ij} = |u_{ij} - R/K|$, and with the thrust coefficient C_T the velocity deficit is

$$\Delta_{ij} = 1 - \frac{s_{\text{down}}}{s_{\text{up}}} = \frac{1 - \sqrt{1 - C_T}}{(1 + K d_{ij}/R)^2}. \quad (1)$$

As in [5], this top-hat wake is tested at the hub center only, so the deficit jumps at the cone edge, and it also reaches a turbine less than R/K upstream of j near its axis (non-physical, kept for comparability). The deficits are combined by the root-sum-square rule,

$$\Delta_i(\theta) = \left(\sum_{j \neq i, \beta_{ij} < \alpha} \Delta_{ij}^2 \right)^{1/2}. \quad (2)$$

As C_T is constant (which overstates the deficits above rated speed), the local wind speed in a direction bin is the free-stream speed times $1 - \Delta_i(\theta)$, a Weibull variable with shape $\zeta(\theta)$ and scale $\psi_i(\theta) = \psi(\theta)[1 - \Delta_i(\theta)]$.

B. Power Model

The power curve of the Kusiak–Song benchmark [5] is $f(s) = 0$ for $s < s_{\text{cut-in}}$, $f(s) = \lambda_1 s + \lambda_0$ for $s_{\text{cut-in}} \leq s \leq s_{\text{rated}}$ and $f(s) = P_{\text{rated}}$ above, with $s_{\text{cut-in}} = 3.5$ m/s, $s_{\text{rated}} = 14$ m/s, $P_{\text{rated}} = 1500$ kW, $\lambda_1 = 140.86$ and $\lambda_0 = -500$. We use it as published: it is slightly negative just above the cut-in speed, jumps from 1472 to 1500 kW at the rated speed and has no cut-out speed. The expected power of turbine i ,

$$E(P_i) = \int_0^{360} p_\theta(\theta) \int_0^\infty f(s) p_s(s; \zeta(\theta), \psi_i(\theta)) ds d\theta, \quad (3)$$

is evaluated, as in [5], by a Riemann sum over N_θ direction bins with probabilities p_l and N_s speed bins between $s_{\text{cut-in}}$ and s_{rated} (Section S-VI-A of the supplementary material). The direction sum keeps the bin width in degrees as a factor; with 24 bins of 15° and $\sum_l p_l = 1$, the benchmark objective $\sum_i E(P_i)$ is thus 15 times the expected farm power in kW, and the gross (wake-only) AEP of the benchmark is $\text{AEP}[\text{MWh}] = (\text{objective}/15) \times 8.76$. For Data Set I, the wake-free benchmark objective of one turbine, $E_{\text{ideal}} = 14,045.74$, corresponds to 936.38 kW (a capacity factor of 62%, far above real sites) or 8,202.7 MWh/yr.

C. Constraint Handling

After every position update, each coordinate is clipped to $[-r, r]$ (box repair; the square, not the circle). The constraints are handled by a penalty on the violations $g_i^b = x_i^2 + y_i^2 - r^2$ and $g_{ij}^s = \ell_{\min} - \ell_{ij}$; all optimizers except MS-SLSQP minimize the penalized wake loss

$$F_p(\mathbf{X}) = \left(NE_{\text{ideal}} - \sum_{i=1}^N E(P_i) \right) + \sum_{i: g_i^b > 0} (1 + \mu g_i^b)^2 + \sum_{i < j: g_{ij}^s > 0} (1 + \mu g_{ij}^s)^2, \quad \mu = 10^{10}, \quad (4)$$

where $\mathbf{X} = (x_1, y_1, \dots, x_N, y_N)$; since NE_{ideal} is the wake-free benchmark objective, the first term is the wake loss reported in our tables. Any violation above about 10^{-7} (m or m^2) makes the penalty exceed the largest possible wake loss, so, up to smaller violations, comparing F_p values is a feasibility-first rule: a feasible layout beats an infeasible one, feasible layouts are compared by wake loss, and infeasible ones by total penalty. MS-SLSQP instead minimizes the wake loss under explicit constraints (Section V). A final layout is counted as feasible if $\ell_{ij} \geq \ell_{\min}$ and $\sqrt{x_i^2 + y_i^2} \leq r$ hold within 10^{-6} m (Section V).

III. PRELIMINARIES: PHASE-1 METHODS AND BASIC VNS

A population method keeps N_p candidate vectors $\mathbf{X}^i \in \mathbb{R}^n$ with components X_m^i and bounds lb_m, ub_m ; unless stated otherwise, one iteration costs N_p objective evaluations.

A. PSO and Its Stability Conditions

We use the canonical global-best PSO [25] in the inertia-weight form [36] (the standard PSO of [37] uses a ring topology instead). Particle i has a position $\mathbf{X}^i$, a velocity $\mathbf{V}^i$ and a personal best $\mathbf{P}^i$, the swarm keeps the global best $\mathbf{G}$, and a particle moves by

$$\begin{aligned} \mathbf{V}^i &\leftarrow w \mathbf{V}^i + c_1 \rho_1 \circ (\mathbf{P}^i - \mathbf{X}^i) + c_2 \rho_2 \circ (\mathbf{G} - \mathbf{X}^i), \\ \mathbf{X}^i &\leftarrow \mathbf{X}^i + \mathbf{V}^i, \end{aligned} \quad (5)$$

where ρ_1, ρ_2 are vectors of independent $U(0, 1)$ numbers, drawn anew for every move, and $\circ$ is the componentwise product. The constriction PSO of Clerc and Kennedy [33] with their standard choice $\varphi_1 = \varphi_2 = 2.05$ (constriction factor $\kappa = 0.7298$) is identical to (5) with

$$w = 0.7298, \quad c_1 = c_2 = \kappa \varphi_1 = 1.49618. \quad (7)$$

PSO denotes (5), (6) with these coefficients, fixed a priori from the literature, and the *old setting* the same algorithm with $w = 0.7$, $c_1 = c_2 = 2$, i.e., the acceleration coefficients of the original PSO [25] combined with an inertia weight. Velocities start at zero and are not clamped; positions are clipped to the bounding square after each move, and a clipped coordinate keeps its velocity (no reflection or damping); $\mathbf{P}^i$ and $\mathbf{G}$ are updated after each evaluation.

Under stagnation (fixed $\mathbf{P}^i$ and $\mathbf{G}$) and without bounds, the expected positions converge (order-1 stability) if $|w| < 1$ and $0 < c_1 + c_2 < 4(1 + w)$, and their variance converges as well (order-2 stability) only if also [23]

$$c_1 + c_2 < \frac{24(1 - w^2)}{7 - 5w}, \quad (8)$$

which also holds under a weaker, non-stagnant assumption [24]. The constriction setting satisfies (8) ($c_1 + c_2 = 2.99 < 3.35$). The old setting is order-1 stable ($4 < 6.8$) but violates the order-2 condition (8) ($4 > 3.50$), so the position variance does not converge. With velocities kept at clipping, the particles then oscillate widely and are often clipped to the square, whose corners lie outside the farm; this plausibly explains the low feasibility of this setting. Section VI-A shows

that, at 6,030 evaluations, this changes the ranking of the compared methods. Likewise, the DE scale factor 0.5 (Section V) exceeds Zaharie's critical value $\sqrt{(1 - \text{CR}/2)/N_p} = 0.14$ (CR = 0.9) [38].

B. Salp Swarm Algorithm and Its Laplacian Variant

The SSA [30] keeps the best solution found so far as the food source $\mathbf{H}$ and splits the sorted population into leaders and followers. A leader $i \leq N_p/2$ updates each coordinate by

$$X_m^i = H_m \pm \xi_1((ub_m - lb_m)\xi_2 + lb_m), \quad (9)$$

with either sign with probability 1/2, $\xi_2 \sim U(0, 1)$ and $\xi_1 = 2 \exp[-(4t/T)^2]$ decreasing with the iteration t out of T ; a follower $i > N_p/2$ moves to the midpoint of its position and the new position of the preceding salp. The leaders thus perturb $\mathbf{H}$ uniformly with a shrinking radius, and the followers only average [31]. LX-SSA [32] gives each follower a second candidate $Y_m^i = X_m^i + \gamma(H_m - X_m^i)$ with the Laplace step [39]

$$\gamma = \phi - \chi \operatorname{sgn}(z - \tfrac{1}{2}) \ln(1 - 2|z - \tfrac{1}{2}|), \quad z \sim U(0, 1), \quad (10)$$

$\phi = 0$, $\chi = 1$, and keeps the better candidate. As published, it evaluates both candidates of each follower and then re-evaluates the kept ones with the new population, so an iteration costs $2N_p$ evaluations and LX-SSA runs half as many iterations as SSA; comparing the two tests LX-SSA as published, not the Laplace step alone. Pseudocode: Section S-I-B of the supplementary material.

C. Basic Variable Neighborhood Search

The basic VNS [28], [29], in the continuous form of [40], improves one incumbent $\mathbf{X}$ with $k_{\max} = 5$ nested ℓ_∞ neighborhoods $\mathcal{N}_k(\mathbf{X}) = \{\mathbf{X}' : \delta_{k-1} < \|\mathbf{X}' - \mathbf{X}\|_\infty \leq \delta_k\}$, $\delta_k = 0.1kr$, of the whole layout. Shaking draws a point uniformly from $\mathcal{N}_k(\mathbf{X})$ (box-repaired), and a best-improvement local search improves it. As the top-hat wake makes F_p discontinuous, this is a compass search: it evaluates all $2n$ moves $\pm h \mathbf{e}_m$ ($\mathbf{e}_m$ the m th unit vector), moves to the best improving one, and halves h (from $0.05r$) when none improves, until $h < 10^{-3}r$.

IV. TWO-PHASE SWARM-VNS DESIGN AND PSO-VNS

A. Two-Phase Template

All hybrids here follow one sequential (relay) template. For a budget of B evaluations and a split $\omega \in (0, 1)$, Phase 1 starts from the seeded initial population, runs a population method for about ωB evaluations and passes its best layout to Phase 2, the basic VNS, which applies the local search to it and then iterates until B evaluations are used. Only Phase 1 differs: PSO in *PSO-VNS*; SSA or LX-SSA in *SSA-VNS* and *LX-SSA-VNS* (with ξ_1 compressed to the Phase-1 iterations, so, unlike PSO, not the start of a stand-alone run); and $\text{round}(\omega B)$ layouts drawn uniformly in the bounding square, including the initial population, in the control *RS-VNS*. VNS alone is the limiting case in which Phase 1 is the initial population. PSO and SSA run for $T_1 = \text{round}((\omega B - N_p)/N_p)$ Phase-1 iterations, LX-SSA for $\text{round}((\omega B - N_p)/(2N_p))$. A swarm phase is useful

Algorithm 1 PSO-VNS for a prescribed turbine count N

Require: budget B , split ω , N_p , w , c_1 , c_2 , radius r , $k_{\max}$

- 1: Draw and evaluate N_p particles $\mathbf{X}^i$; $\mathbf{V}^i \leftarrow \mathbf{0}$; $\mathbf{P}^i \leftarrow \mathbf{X}^i$;
 $\mathbf{G} \leftarrow \arg \min_i F_p(\mathbf{P}^i)$ ▷ Phase 1
- 2: **for** $t = 1, \dots, T_1$ and $i = 1, \dots, N_p$ **do**
- 3: Move $\mathbf{X}^i$ by (5), (6); box repair; evaluate $F_p(\mathbf{X}^i)$;
 update $\mathbf{P}^i$, $\mathbf{G}$
- 4: **end for**
- 5: $\mathbf{X} \leftarrow \text{LOCALSEARCH}(\mathbf{G})$; $k \leftarrow 1$ ▷ Phase 2
- 6: **while** fewer than B evaluations have been used **do**
- 7: $\mathbf{X}' \leftarrow \text{LOCALSEARCH}(\text{uniform point of } \mathcal{N}_k(\mathbf{X}))$
- 8: **if** $F_p(\mathbf{X}') < F_p(\mathbf{X})$ **then** $\mathbf{X} \leftarrow \mathbf{X}'$, $k \leftarrow 1$ **else** $k \leftarrow$
 $k \bmod k_{\max} + 1$
- 9: **end while**
- 10: **return** best layout evaluated

only if it gives VNS a better start than random sampling with about the same number of evaluations (at $\omega = 0.5$, 3,015 for RS-VNS and $N_p(T_1 + 1) = 3,030$ for the swarms); the component analysis (Section VII) tests this for each Phase-1 choice, with RS-VNS as a weak, no-search minimum bar.

B. PSO-VNS

PSO-VNS (Algorithm 1) is not a new PSO variant: Phase 1 is the plain PSO of Section III-A. With coefficients that satisfy (8), the particles contract around their personal and global bests; empirically (Section VII), this gives VNS a better start. Since the coefficients do not depend on the iteration, Phase 1 reproduces exactly, for a given seed, the first $N_p(T_1 + 1)$ evaluations of a PSO run, so comparing PSO-VNS with PSO isolates the use of the second part of the budget.

PSO-VNS uses $N_p = 30$, the coefficients (7) and the VNS settings above, and adds one parameter, the split ω . With the default $\omega = 0.5$ and $B = 6,030$, PSO runs for $T_1 = 100$ iterations (99.5 rounded up) and VNS receives the remaining 3,000 evaluations. None of these values was tuned; Section VII studies ω .

V. EVALUATION PROTOCOL

A. Cases and Methods

The 68 benchmark cases combine the two wind data sets of the Kusiak–Song benchmark [5], [8] (Table S1 of the supplementary material) with circular farms of radius $r = 500$, 750 and 1000 m and $N = 2, \dots, 10$, $2, \dots, 12$ and $2, \dots, 15$ turbines ($R = 38.5$ m, $\ell_{\min} = 4D = 308$ m, $C_T = 0.8$, $K = 0.075$).

The *main comparison* includes eight methods ($N_p = 30$ where applicable): PSO-VNS (Algorithm 1, $\omega = 0.5$); PSO; SSA-VNS; SSA; LX-SSA; DE/rand/1/bin [41] (scale factor 0.5, crossover rate 0.9); VNS; and MS-SLSQP, a multi-start sequential least-squares programming method (SciPy's SLSQP [42], forward-difference gradients). All methods except MS-SLSQP minimize F_p (4) over $[-r, r]^{2N}$; MS-SLSQP minimizes the wake loss under explicit constraints, restarting from the initial layouts and then from random ones, and returns its best feasible layout (settings: Section S-I). DE and

MS-SLSQP use standard, untuned settings. The *component analysis* (Section VII) adds LX-SSA-VNS and the no-search control RS-VNS (Section IV-A), whose 3,015 random layouts cost about as much as the 3,030 Phase-1 evaluations of the hybrids; a split study varies ω on twelve cases (Section VII).

B. Comparison Design and Statistics

Equal budgets and paired seeds. Unless stated otherwise, a run uses exactly $B = 6,030$ objective evaluations, counted by a wrapper around the objective, including the initial population and the gradient evaluations of MS-SLSQP. Each method uses the seeds 1–30 and draws its initial population first, so runs with the same seed start from the same population and are paired (where logged, the best initial objective agrees across all methods, including the earlier version’s runs).

Feasibility-aware ranking. A final layout counts as feasible if all spacing and boundary constraints hold within 10^{-6} m. Within a case, a method is ranked by the mean benchmark objective of its feasible runs only if at least 15 of its 30 runs (5 of 10 for 10 seeds) are feasible, and otherwise below all such methods, by its number of feasible runs. This limits, but does not remove, survivorship bias: a method with 15 to 29 feasible runs is still ranked by its surviving runs.

Statistics. All tests are two-sided ($\alpha = 0.05$), with Holm’s correction [43] within each family of comparisons. 1) Case ranks: the Friedman test with the Iman–Davenport F test and post hoc tests of the average ranks against PSO-VNS [26]. 2) Case means (primary tests): since mean-rank post hoc tests depend on the pool [27], Wilcoxon signed-rank tests of PSO-VNS against each method on the per-case mean wake losses; a case in which only one of the two methods has at least 15 feasible runs counts as a maximal difference in its favor (2 for SSA, 2 for LX-SSA and 18 for DE cases), and one in which neither has is dropped (0 cases). Headline mean differences of case means (cases in which both qualify) carry 95% bootstrap confidence intervals over cases. Case-mean p values quoted in the text for single contrasts are unadjusted; the tables give Holm-adjusted values, and every conclusion drawn holds under both. 3) Runs: per case, Wilcoxon signed-rank tests on the 30 seed-paired runs (infeasible runs below all feasible ones, ordered by minimum spacing), Holm-corrected over the comparisons of the case, with the matched-pairs rank-biserial correlation r_{rb} (zero differences dropped). The counts of significant wins, ties and significant losses (W/T/L) are descriptive: Holm acts only within a case, so a few significant cases can arise by chance, and a pair’s counts differ between tables with different families. The 68 cases are six site–wind clusters (data set, radius) with nested N , not independent draws; the p values describe this benchmark, not a population of farms, and subgroup splits are exploratory.

C. Tests Beyond the Benchmark and Reproducibility

Initialization and budget. Random initial layouts are usually infeasible; the feasible initialization draws the initial layouts inside the site from the seeded generator and separates the turbines with L-BFGS-B on a packing penalty, without objective

evaluations (Section S-I). The nine methods other than RS-VNS, which would need a packing solve per random layout, use it at 6,030 evaluations; with random initialization, all ten methods run at 6,030, 30,030 and 120,030 evaluations. Both studies use the six largest cases (largest N of each farm and data set) and the Horns Rev 1 block, with 30 seeds (10 for Horns Rev 1 at 120,030 evaluations).

Other sites and models. The Horns Rev 1 block is PyWake’s 16-turbine example [44]: V80 turbines ($D = 80$ m), the measured 12-sector wind climate, the parallelogram through the corner turbines as boundary (Section VIII-B), $\ell_{\min} = 4D$ and the Jensen wake with $K = 0.04$. The 16- and 36-turbine scenarios of IEA37 Case Study 1 [20], [45] (circular sites of radius 1300 and 2000 m, $\ell_{\min} = 2D$, 16 directions at one speed, a simplified Bastankhah Gaussian wake) test optimizers rather than represent a site; all ten methods run on them at 6,030 and 30,030 evaluations (30 seeds), and their layouts are compared with the submitted ones.

Verification and code. Our objective matches the original implementation (largest absolute difference 8.7×10^{-11} objective units over 720 archived final objectives) and the official IEA37 calculator (relative difference below 10^{-11}). Our reused code of SSA, LX-SSA, PSO and DE (also in the hybrids) reproduces the result distributions of the earlier version (2 of 24 Mann–Whitney tests reject at $\alpha = 0.05$, about the 1.2 expected by chance). The runs used Linux containers (Intel Xeon, 2.1 GHz, 4 virtual cores; Python 3.11.15, NumPy 2.4.6, SciPy 1.17.1); code, per-run results and scripts are public (Data Availability).

VI. RESULTS ON THE BENCHMARK

This section reports the 68 benchmark cases (20,400 runs, 6,030 evaluations each); per-case means and standard deviations are given in Tables S3–S8 of the supplementary material.

A. Effect of the Baseline Setting

The old setting violates the order-2 (variance) stability condition (8). We ran PSO with it on the same cases with the same budget and seeds. Compared with the constriction setting (7), it returns a feasible layout in only 86.0% of the runs (fewer than 15 feasible runs in 8 cases), is significantly worse in 56 of the 68 cases and never better (Wilcoxon, 30 seed-paired runs), and has a mean wake loss 0.61 percentage points higher over the 60 cases in which both settings have at least 15 feasible runs. As (8) assumes stagnation and unbounded search, the low feasibility plausibly comes from the interplay of the old setting with the clipping to the square, after which velocities are kept (Section III-A). In the pool of Table II, the old setting would place PSO fifth (average rank 5.12) instead of second; with the constriction setting, PSO ranks ahead of SSA, LX-SSA and SSA-VNS.

Table I shows the effect on a comparison: it ranks the methods of Table II, with LX-SSA-VNS in place of PSO-VNS, once with each PSO setting; all other methods use identical runs in both columns, so their rank changes come only from PSO. With the old setting, SSA-VNS has the best average rank, PSO ranks behind a salp-swarm hybrid, and the W/T/L

TABLE I

EFFECT OF THE PSO SETTING ON THE METHOD POOL OF THE OLD-SETTING COMPARISON (THE POOL OF TABLE II WITH LX-SSA-VNS INSTEAD OF PSO-VNS; 68 CASES, 6,030 EVALUATIONS): AVERAGE RANK (1 = BEST) WITH THE OLD ($w = 0.7$, $c_1 = c_2 = 2$) AND THE CONSTRICTION SETTING, AND RUN-LEVEL W/T/L OF THE SALP-SWARM HYBRIDS AGAINST PSO

Method	PSO setting	
	Old	Constriction
LX-SSA-VNS	2.65	3.41
SSA-VNS	1.99	2.77
LX-SSA	5.65	6.06
SSA	4.81	5.34
PSO	5.04	1.53
DE	6.99	7.03
VNS	3.59	4.15
MS-SLSQP	5.28	5.71
PSO position	5	1
LX-SSA-VNS vs. PSO (W/T/L)	35/33/0	0/21/47
SSA-VNS vs. PSO (W/T/L)	41/27/0	0/22/46

Ranks: feasibility-aware rule (fewer than 15 of 30 feasible runs in a case = ranked last); bold: best average rank. W/T/L: cases in which the hybrid is significantly better / not different / worse than PSO (two-sided Wilcoxon signed-rank test, 30 seed-paired runs, Holm-adjusted over the seven comparisons of the hybrid in each case, $\alpha = 0.05$).

of LX-SSA-VNS and SSA-VNS against PSO are 35/33/0 and 41/27/0. With the constriction setting, PSO ranks ahead of every salp-swarm method, and these W/T/L become 0/21/47 and 0/22/46. Changing the coefficients of one baseline thus changes the ranking: at 6,030 evaluations, PSO moves from behind to ahead of the salp-swarm hybrids. The constriction coefficients are standard values [33], not tuned on these cases; the other methods keep standard, untuned settings. The old setting was run only at 6,030 evaluations, and with more evaluations the order changes again: on the six largest cases at 120,030 evaluations, SSA-VNS ranks ahead of PSO (average ranks 4.17 and 5.83; Section VIII-A). This is a property of the setting, not of any particular study: on this benchmark, at 6,030 evaluations and with clipping, a PSO baseline that violates (8) performs markedly worse, so a method compared with it can appear stronger than it is. Under these conditions the salp-swarm methods, including our own LX-SSA, do not outperform a correctly configured PSO, which is why PSO forms the first phase of PSO-VNS.

B. Overall Comparison

Over the eight methods (Table II), the Friedman test rejects the hypothesis of equal performance ($\chi_F^2 = 319.3$, $p = 4.6 \times 10^{-65}$; Iman-Davenport $F_F = 136.5$, $p = 6.7 \times 10^{-109}$). PSO-VNS has the best average rank (1.82) and alone ranks first in 29 cases. In the Holm-adjusted post hoc tests (p_z in Table II; tied ranks make them conservative), PSO-VNS ranks significantly better than SSA-VNS, VNS, SSA, MS-SLSQP, LX-SSA and DE, but not than PSO ($p_{\text{Holm}} = 0.71$). $\overline{\Delta L}$ averages only over the cases in which both methods have at least 15 feasible runs (fewer than 68 for SSA, LX-SSA and DE).

The W/T/L counts of the run-level tests (Table III) are descriptive; Holm's correction is applied within each case over

TABLE II

CASE-LEVEL ANALYSIS OVER THE 68 BENCHMARK CASES. AVG. RANK: AVERAGE RANK OF THE MEAN FEASIBLE OBJECTIVE (1 = BEST; METHODS WITH FEWER THAN 15 FEASIBLE RUNS IN A CASE RANKED LAST); "BEST": CASES IN WHICH THE METHOD ALONE RANKS FIRST; "FEAS.": FEASIBLE RUNS (%); p_z : HOLM-ADJUSTED p OF THE AVERAGE-RANK TEST AGAINST PSO-VNS. BECAUSE MEAN-RANK TESTS DEPEND ON THE POOL OF METHODS [27], p_W AND $\overline{\Delta L}$ GIVE THE TWO-SIDED WILCOXON SIGNED-RANK TEST ON THE 68 PER-CASE MEAN WAKE LOSSES (HOLM-ADJUSTED OVER THE 7 METHODS) AND THE MEAN WAKE-LOSS DIFFERENCE OVER THE CASES IN WHICH BOTH METHODS HAVE AT LEAST 15 FEASIBLE RUNS (66 FOR SSA, 66 FOR LX-SSA, 50 FOR DE, OTHERWISE 68; PERCENTAGE POINTS, PSO-VNS MINUS METHOD; NEGATIVE = PSO-VNS BETTER)

Method	Avg. rank	Best	Feas.	p_z	p_W	$\overline{\Delta L}$
PSO-VNS	1.82	29	100.0	—	—	—
PSO	1.97	25	100.0	0.713	0.296	-0.018
SSA-VNS	3.45	0	99.6	2.0×10^{-4}	5.7×10^{-11}	-0.330
VNS	4.40	3	100.0	2.4×10^{-9}	1.3×10^{-10}	-0.465
SSA	5.38	0	97.6	8.3×10^{-17}	4.5×10^{-11}	-0.611
MS-SLSQP	5.88	1	100.0	1.8×10^{-21}	4.2×10^{-11}	-0.435
LX-SSA	6.06	0	96.9	3.3×10^{-23}	4.5×10^{-11}	-0.765
DE	7.04	0	71.4	1.0×10^{-34}	5.6×10^{-11}	-0.887

Friedman $\chi_F^2 = 319.3$ (7 d.f.), $p = 4.6 \times 10^{-65}$;

Iman-Davenport $F_F = 136.5$ (7, 469 d.f.), $p = 6.7 \times 10^{-109}$

TABLE III

PAIRWISE OUTCOME OF PSO-VNS AGAINST EACH METHOD: CASES IN WHICH PSO-VNS IS SIGNIFICANTLY BETTER / NOT DIFFERENT / WORSE (TWO-SIDED WILCOXON SIGNED-RANK TEST, 30 SEED-PAIRED RUNS, HOLM-ADJUSTED OVER THE SEVEN COMPARISONS OF PSO-VNS IN EACH CASE, A DIFFERENT FAMILY FROM TABLE IV, $\alpha = 0.05$; AN INFEASIBLE RUN RANKS BELOW EVERY FEASIBLE RUN); LAST ROW: MEDIAN RANK-BISERIAL CORRELATION (POSITIVE = PSO-VNS BETTER; ZERO DIFFERENCES DROPPED, $r_{\text{rb}} = 0$ IF ALL ARE ZERO)

DS	r (m)	Cases	PSO	SSA-VNS	SSA	LX-SSA	DE	VNS	MS-SLSQP
I	500	9	0/8/1	6/3/0	8/1/0	8/1/0	7/2/0	6/3/0	7/2/0
I	750	11	1/8/2	9/2/0	10/1/0	10/1/0	9/2/0	8/3/0	10/1/0
I	1000	14	3/11/0	9/5/0	12/2/0	12/2/0	12/2/0	12/2/0	12/2/0
II	500	9	3/5/1	7/2/0	7/2/0	8/1/0	7/2/0	7/1/1	8/1/0
II	750	11	2/9/0	8/3/0	10/1/0	10/1/0	9/2/0	8/3/0	9/2/0
II	1000	14	5/9/0	10/4/0	12/2/0	12/2/0	12/2/0	10/4/0	13/1/0
All		68	14/50/4	49/19/0	59/9/0	60/8/0	56/12/0	51/16/1	59/9/0
$\bar{r}_{\text{rb}}$			+0.08	+0.72	+0.98	+0.97	+1.00	+0.84	+0.99

the seven comparisons with PSO-VNS (the component analysis uses eleven contrasts and so gives slightly different counts, Section VII). Against PSO, PSO-VNS is significantly better in 14 cases, tied in 50 and worse in 4. The mean difference of the case means, -0.018 percentage points (95% bootstrap CI [-0.045, 0.008]), is not significant ($p_{\text{Holm}} = 0.30$), and PSO has the lower case mean in 26 of the 68 cases. In an exploratory, post hoc split, the advantage of PSO-VNS is concentrated in Data Set II with $N \geq 10$, where it has the lower mean wake loss in 10 of 10 cases (unadjusted Wilcoxon $p = 0.002$; significant run-level wins/losses 8/0). In Data Set I with $N \geq 10$, PSO has the lower mean wake loss in 6 of 10 cases (wins/losses of PSO-VNS 1/2), and for $N < 10$ both reach nearly the same wake loss (mean absolute difference 0.03 percentage points). Against SSA-VNS, the same design with a salp-swarm first phase, PSO-VNS is significantly better in 49 cases and never worse ($p_{\text{Holm}} = 5.7 \times 10^{-11}$). Against SSA, LX-SSA, DE, VNS and MS-SLSQP, it is significantly

better in most cases; it is never significantly worse than SSA-VNS, SSA, LX-SSA, DE and MS-SLSQP and significantly worse than VNS in one case. PSO-VNS is thus a simple reference method with limits: it has the best average rank and is feasible in every run, but is not significantly better than PSO. On the six largest cases it has the best average rank at all three budgets (exploratory, no test; first in 3 of the 6 cases at 120,030 evaluations; Section VIII-A). The order of the leading methods depends on the test conditions: with feasible initial layouts, VNS has the best average rank (Section VIII-A); when the final layouts are re-evaluated with a Gaussian wake model, PSO ranks first, narrowly ahead of PSO-VNS (Section IX); and on IEA37 Case Study 1, the best layouts remain below the best published ones (Section VIII-C).

C. Solution Quality, Feasibility and Convergence

The wake loss grows with N , falls as the farm becomes larger, and is higher for the broader wind rose of Data Set II (Section S-II-D of the supplementary material). Over the six largest cases, the mean wake loss of PSO minus that of PSO-VNS ranges from -0.18 to $+0.41$ percentage points, with PSO-VNS lower in four of them. PSO-VNS ends with a feasible layout in all runs (other methods: Table II; DE 71.4%). The feasibility counts of PSO, SSA, LX-SSA and DE are the same for both wind data sets, because as long as every candidate is infeasible, the penalty dominates F_p . In the densest cases (500 m, $N = 10$), the global best of the PSO phase is feasible at the switch in only 63.3% of the runs, but all runs are feasible after the VNS phase.

Fig. 1 shows the second phase directly: PSO-VNS and PSO both keep improving after the switch. Over all feasible runs, the VNS phase removes on average 30.3% (median 22.6%) of the wake loss left at the end of the PSO phase and turns 26 runs that were infeasible at the switch into feasible ones, whereas continuing PSO for the same 3,000 evaluations removes 29.2% (median 20.6%).

VII. COMPONENT ANALYSIS OF TWO-PHASE HYBRIDS

The eight variants of Table IV combine a Phase 1 (PSO, SSA, LX-SSA, random sampling or none) with or without the same VNS phase, at 6,030 evaluations and with the same seeds (convergence in Fig. 1); their average ranks differ (Friedman $\chi^2_F = 326.8$, 7 d.f., $p = 1.1 \times 10^{-66}$). Holm's correction runs over the eleven contrasts of a case, not the seven comparisons of Table III, so PSO-VNS vs. PSO is 8/58/2 here and 14/50/4 there. Case-level p values are from the Wilcoxon test on the 68 case means (Section V); $\overline{\Delta L}$ is the mean difference of the case means (percentage points), with its 95% bootstrap interval over the cases.

VNS phase. The VNS phase significantly improves both salp-swarm methods: SSA-VNS is 39/29/0 (W/T/L) against SSA and LX-SSA-VNS 35/33/0 against LX-SSA ($\overline{\Delta L} = -0.328$ and -0.431 ; these contrasts also compress the ξ_1 schedule into Phase 1). The PSO contrast is clean, since PSO-VNS and PSO share their first 3,030 evaluations (Section IV): switching to VNS instead of continuing the swarm gives a

TABLE IV
COMPONENT ANALYSIS OVER THE 68 BENCHMARK CASES (30 SEED-PAIRED RUNS, 6,030 EVALUATIONS). TOP: AVERAGE RANK AMONG THE EIGHT VARIANTS (FRIEDMAN $\chi^2_F = 326.8$, 7 D.F., $p = 1.14 \times 10^{-66}$) AND FEASIBLE RUNS. BOTTOM: PLANNED CONTRASTS; W/T/L: CASES IN WHICH THE FIRST VARIANT IS SIGNIFICANTLY BETTER / NOT DIFFERENT / WORSE (RUN-LEVEL WILCOXON, HOLM-ADJUSTED OVER THE ELEVEN CONTRASTS OF EACH CASE, A DIFFERENT FAMILY FROM TABLE III, $\alpha = 0.05$); p_W : WILCOXON TEST ON THE 68 PER-CASE MEAN WAKE LOSSES, HOLM-ADJUSTED OVER THE ELEVEN CONTRASTS; $\overline{\Delta L}$: MEAN DIFFERENCE OF THE CASE-MEAN WAKE LOSSES (PERCENTAGE POINTS; NEGATIVE = FIRST VARIANT BETTER)

Variant	Phase 1	Phase 2	Avg. rank	Feas. (%)
PSO-VNS	PSO	VNS	1.87	100.0
SSA-VNS	SSA	VNS	3.79	99.6
LX-SSA-VNS	LX-SSA	VNS	4.62	99.9
RS-VNS	random sampling	VNS	4.46	99.4
VNS	–	VNS	5.32	100.0
PSO	PSO	–	2.10	100.0
SSA	SSA	–	6.56	97.6
LX-SSA	LX-SSA	–	7.29	96.9

Contrast	Isolates	W/T/L	p_W	$\overline{\Delta L}$
PSO-VNS vs. PSO	VNS phase (PSO)	8/58/2	0.592	−0.018
SSA-VNS vs. SSA	VNS phase (SSA)	39/29/0	4.0×10^{-10}	−0.294
LX-SSA-VNS vs. LX-SSA	VNS phase (LX-SSA)	35/33/0	8.3×10^{-11}	−0.394
PSO-VNS vs. VNS	swarm start vs. best initial point	49/18/1	4.0×10^{-10}	−0.465
PSO-VNS vs. RS-VNS	PSO vs. random sampling	48/20/0	2.1×10^{-10}	−0.399
SSA-VNS vs. RS-VNS	SSA vs. random sampling	2/66/0	0.001	−0.068
LX-SSA-VNS vs. RS-VNS	LX-SSA vs. random sampling	1/67/0	0.718	−0.005
PSO-VNS vs. SSA-VNS	PSO vs. SSA as Phase 1	46/22/0	1.7×10^{-10}	−0.330
PSO-VNS vs. LX-SSA-VNS	PSO vs. LX-SSA as Phase 1	48/20/0	1.6×10^{-10}	−0.393
SSA-VNS vs. LX-SSA-VNS	Laplace step in the hybrid	2/66/0	2.2×10^{-5}	−0.063
LX-SSA vs. SSA	Laplace step alone	0/62/6	9.3×10^{-7}	+0.153

small gain ($\overline{\Delta L} = -0.018$) that is not significant over all 68 cases ($p = 0.30$; Section VI-B).

Swarm phase. Two controls isolate Phase 1: VNS alone, and RS-VNS, whose Phase 1 samples uniform random layouts in the bounding square (3,015 evaluations, about as many as the 3,030 of the swarms). RS-VNS is a weak, no-search control: all N turbines of a uniform sample lie inside the circular site only with probability $(\pi/4)^N$, so its best sample is often just the least infeasible one (feasible at the switch in 66.7% of the runs), and beating RS-VNS is a minimum requirement for a swarm phase. A PSO phase clearly meets it: PSO-VNS is 48/20/0 against RS-VNS ($\overline{\Delta L} = -0.399$ [−0.495, −0.308]) and 49/18/1 against VNS ($\overline{\Delta L} = -0.465$). An SSA phase gives only a small gain: SSA-VNS and RS-VNS are mostly tied at the run level (2/66/0), and on the case means SSA-VNS is better by far less than PSO-VNS ($\overline{\Delta L} = -0.068$ [−0.108, −0.034], $p = 4.5 \times 10^{-4}$). An LX-SSA phase gives no gain (LX-SSA-VNS vs. RS-VNS: 1/67/0, $p = 0.72$).

Laplace step. The Laplace step makes the salp swarm worse. Without VNS, LX-SSA is 0/62/6 against SSA ($\overline{\Delta L} = +0.166$, $p = 1.9 \times 10^{-7}$); in the hybrid, SSA-VNS is 2/66/0 against LX-SSA-VNS and better on the case means ($\overline{\Delta L} = -0.063$, $p = 5.4 \times 10^{-6}$). As published, LX-SSA costs $2N_p$ evaluations

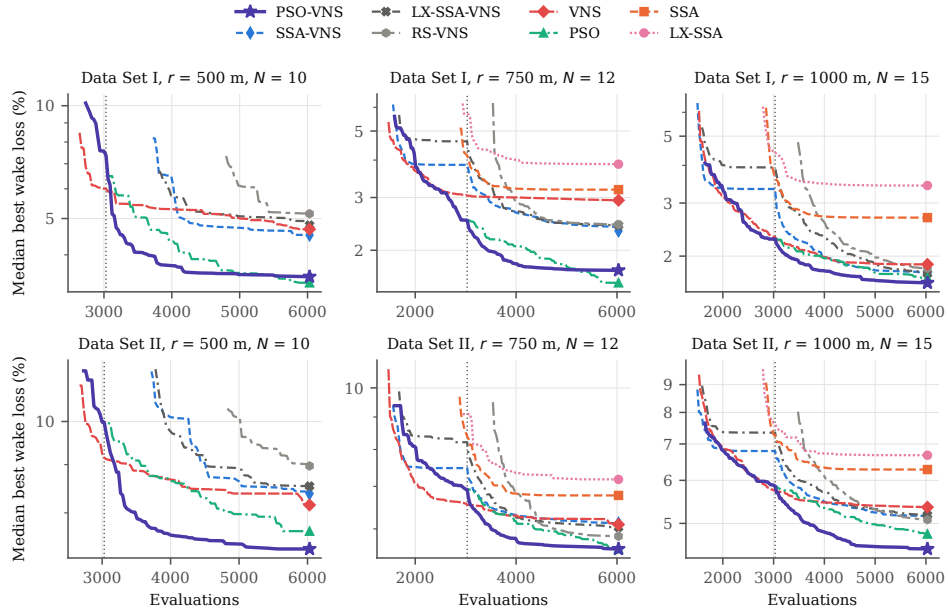


Fig. 1. Convergence of the eight component-analysis variants (Table IV) for the largest turbine count of each farm: median best feasible wake loss over 30 runs versus evaluations. PSO-VNS and PSO coincide up to 3,030 evaluations (dotted line), where PSO-VNS switches to VNS. Curves of the main comparison: Fig. S3 of the supplementary material.

TABLE V

BUDGET SPLIT OF PSO-VNS: SHARE ω OF THE 6,030 EVALUATIONS FOR PSO ($\omega = 0.25, 0.5, 0.75; \omega = 1$: PSO ALONE, SAME SEEDS) ON 12 CASES (MIDDLE AND LARGEST N OF EACH FARM AND DATA SET, 30 SEEDS): MEAN WAKE LOSS, AVERAGE RANK AMONG THE FOUR SETTINGS, AND CASES IN WHICH THE DEFAULT $\omega = 0.5$ IS SIGNIFICANTLY BETTER / NOT DIFFERENT / WORSE (RUN-LEVEL WILCOXON, HOLM-ADJUSTED OVER THE 4 COMPARISONS OF EACH CASE)

ω	Mean loss (%)	Avg. rank	0.5 vs. ω (W/T/L)
0.25	2.796	3.75	6/6/0
0.5	2.643	2.42	—
0.75	2.543	1.08	0/8/4
1 (PSO alone)	2.693	2.75	3/8/1

$\omega = 0.75$ vs. $\omega = 1$ (PSO alone): 10/2/0 (W/T/L from the $\omega = 0.75$ side)

per iteration (Section III-B) and so runs half as many iterations as SSA; these contrasts do not separate the Laplace step from this cost.

Choice of Phase 1. With the same VNS phase, PSO-VNS is 46/22/0 against SSA-VNS and 48/20/0 against LX-SSA-VNS. At the switch, its global best is feasible more often (98.7% of the runs vs. 95.7% and 94.1%) and, when feasible, has a lower wake loss (1.63% vs. 2.01% and 2.11%), so VNS starts from a better layout. These data do not show whether PSO helps mainly by reaching feasibility sooner or by exploring the wake-loss landscape better.

Budget split. Table V compares the split $\omega = 0.5$, fixed before the experiments and used for all other results, with $\omega = 0.25, 0.75$ and the end point $\omega = 1$ (plain PSO) on the twelve cases with the middle and the largest N of each farm (per-case values: Table S13 of the supplementary material). With $\omega = 0.75$, the mean wake loss is lower than with $\omega = 0.5$ in eleven of the twelve cases, significantly in four, and

0.10 percentage points lower on average, and the default is significantly better in no case; $\omega = 0.25$ is significantly worse than the default in six cases and significantly better in no case. Plain PSO has a higher mean loss than $\omega = 0.75$ (2.693% vs. 2.543%, average rank 2.75; $\omega = 0.75$ against it: 10/2/0), so the best tested split is not the PSO end point. As $\omega = 0.75$ is only the best of four values on twelve test cases, it is a candidate setting to be confirmed on the target problem; we do not re-tune ω for the main comparison, which may slightly understate PSO-VNS.

VIII. TESTS BEYOND THE BENCHMARK

This section tests the conclusions of Section VI at larger budgets, with feasible starts, on a real site and on an external benchmark.

A. Budget Scaling and Feasibility-Preserving Initialization

Budget. On the six largest benchmark cases, PSO-VNS keeps the best average rank at all three budgets (6,030, 30,030 and 120,030 evaluations: 1.50, 1.17 and 2.17; Table VI; Fig. S9 of the supplementary material). This is exploratory (six cases, no test of the rank differences), and PSO-VNS ranks first in 4, 5 and 3 of the six cases. The mean wake loss of PSO minus that of PSO-VNS is +0.10, +0.18 and +0.27 percentage points (per case: Table S15). The order of the other methods depends on the budget. PSO falls back from an average rank of 1.83 to 5.83, so that at 120,030 evaluations SSA-VNS ranks ahead of it (4.17), while VNS (6.50 to 3.17) and DE move up. At 120,030 evaluations, VNS, SSA-VNS, RS-VNS, LX-SSA-VNS and DE all come within 0.17 percentage points of the mean wake loss of PSO-VNS (3.14%).

TABLE VI

SIX LARGEST BENCHMARK CASES: MEAN WAKE LOSS (%) AND FEASIBLE RUNS (%) AT 6,030 EVALUATIONS WITH RANDOM AND FEASIBILITY-PRESERVING INITIALIZATION, AND AVERAGE RANK AT 6,030, 30,030 AND 120,030 EVALUATIONS (RANDOM INITIALIZATION; 30 SEEDS); N/R: NOT RUN

Method	Random init.		Feasible init.		Avg. rank		
	Loss	Feas.	Loss	Feas.	6,030	30,030	120,030
PSO-VNS	4.084	100.0	3.938	100.0	1.50	1.17	2.17
SSA-VNS	4.678	95.0	3.955	100.0	4.33	4.17	4.17
LX-SSA-VNS	4.802	98.9	3.952	100.0	5.17	5.67	5.83
RS-VNS	4.830	93.3	n/r	n/r	4.67	6.33	4.67
VNS	4.950	100.0	3.810	100.0	6.50	4.83	3.17
PSO	4.181	100.0	5.549	100.0	1.83	2.50	5.83
SSA	4.828	80.0	4.947	100.0	8.33	8.67	8.83
LX-SSA	5.379	77.8	5.048	100.0	8.67	9.83	10.00
DE	—	1.1	5.549	100.0	10.00	5.17	3.33
MS-SLSQP	4.522	99.4	4.294	100.0	4.00	6.67	7.00

TABLE VII

HORNS REV 1 16-TURBINE BLOCK (WAKE-FREE AEP 148.81 GWh/YR): MEAN AEP (GWh/YR) OVER THE FEASIBLE RUNS OF 30 SEEDS AT 6,030 EVALUATIONS, FEASIBLE RUNS, HOLM-ADJUSTED WILCOXON p OF PSO-VNS VS. EACH METHOD, AND MEAN AEP LOSS (%) AT 6,030 EVALUATIONS WITH RANDOM (R) AND FEASIBILITY-PRESERVING (F) INITIALIZATION AND AT 30,030 AND 120,030 EVALUATIONS (R; 10 SEEDS AT 120,030); SUPERScript: FEASIBLE RUNS WHEN NOT ALL RUNS ARE FEASIBLE

Layout / method	6,030 evaluations, R			Loss (%)			
	AEP	Feas.	p_{Holm}	6k R	6k F	30k	120k
Installed layout	139.82	—	—	6.04	—	—	—
PSO-VNS	138.29	30/30	—	7.07	6.97	6.52	6.40
SSA-VNS	137.59	30/30	2.6×10^{-4}	7.54	6.98	7.19	6.47
LX-SSA-VNS	137.41	30/30	1.6×10^{-4}	7.66	7.02	7.07	6.44
RS-VNS	137.58	28/30	1.1×10^{-4}	7.54 ²⁸	n/r	7.04	6.25
VNS	137.35	30/30	1.1×10^{-4}	7.70	6.91	6.85	6.23
PSO	137.79	19/30	4.6×10^{-5}	7.40 ¹⁹	8.35	6.78 ²⁸	6.63
SSA	136.25	30/30	2.1×10^{-7}	8.44	7.83	8.17	7.62
LX-SSA	136.00	28/30	6.5×10^{-8}	8.61 ²⁸	7.89	8.50	8.28
DE	—	0/30	1.7×10^{-8}	—	8.35	7.40 ⁸	6.12
MS-SLSQP	136.31	30/30	1.7×10^{-8}	8.40	8.00	7.82	7.31

Feasible starts. With feasibility-preserving initialization, the share of feasible runs rises for SSA-VNS (95.0% to 100.0%), SSA (80.0% to 100.0%), LX-SSA (77.8% to 100.0%) and DE (1.1% to 100.0%). VNS alone is then best, with the lowest mean wake loss and the best average rank; PSO-VNS has an average rank of 2.83 and a mean wake loss of 3.94% (4.08% with random starts; Table S14). PSO and DE never improve on the best initial layout in these runs. As the turbine labels are arbitrary, velocity and difference moves that combine two feasible layouts typically make turbines overlap; the feasibility-first comparison rejects them, and the global best of PSO does not move. The single-coordinate moves of VNS are not affected. From random starts, the value of PSO as Phase 1 may therefore lie mainly in reaching feasibility under the penalty rather than in a better search of the wake-loss landscape, and the advantage of PSO-VNS depends on the initialization.

B. Horns Rev 1 16-Turbine Block

The Jensen wake of this model uses a hub-center in-wake test, root-sum-square superposition and C_T at the free-

TABLE VIII

IEA37 CASE STUDY 1, 16- AND 36-TURBINE SCENARIOS: AEP IN MWh OF THE BEST RUN OF EACH METHOD AT 6,030 AND 30,030 EVALUATIONS (30 SEEDS), COMPARED WITH THE EXAMPLE LAYOUT AND THE BEST FEASIBLE PUBLISHED LAYOUT [20], [45]

Method / source	16 turbines ($r = 1300$ m)		36 turbines ($r = 2000$ m)	
	6,030	30,030	6,030	30,030
Example layout	366,941.6		737,883.1	
Best feasible published ^a	418,924.4		863,676.3	
PSO-VNS	402,782.4	413,331.9	787,708.9	827,385.0
PSO	403,753.1	412,702.4	787,988.7	830,256.4
SSA-VNS	403,251.3	406,924.6	766,803.3	819,930.5
SSA	395,150.2	401,459.3	750,707.4	802,538.2
LX-SSA	393,284.6	398,770.0	746,301.3	789,311.3
DE	357,391.3	397,402.7	—	—
VNS	400,169.3	404,076.0	785,791.3	824,230.2
MS-SLSQP	405,433.8	410,349.5	748,571.1	833,365.0

^a Participant 4 in both scenarios (SNOPT with wake expansion continuation [TBD: verify against Baker *et al.* (2019)]). A higher submitted AEP places turbines up to 3.5 m outside the boundary and is not counted.

stream speed, with 5° direction bins (six per sector) and 1-m/s speed bins; the parallelogram is enlarged by 0.1% so that the installed turbines lie inside it, the boundary violation in (4) is the distance (m) outside it, and the search box has a half-width of 943 m (also the r of the VNS radii). For the 80-turbine farm, our AEP agrees with PyWake [44] within 0.64%. The installed layout, a regular $7D$ grid, reaches 139.82 GWh/yr (wake loss 6.04%; Table VII; layouts: Fig. S10 of the supplementary material). At 6,030 evaluations, PSO-VNS is feasible in 30/30 runs and has the highest mean AEP, 138.29 GWh/yr (SD 0.75); in the seed-paired run-level test (infeasible runs ranked last) it is significantly better than every other method ($p_{\text{Holm}} \leq 2.6 \times 10^{-4}$). Its best run reaches 139.78 GWh/yr, and 0 of its 30 runs exceed the installed layout. At 30,030 evaluations, PSO-VNS still has the highest mean AEP, 139.11 GWh/yr (loss 6.52%), and 3 of its 30 runs exceed the installed layout. At 120,030 evaluations (10 seeds), its loss of 6.40% is no longer the lowest; DE has the highest mean AEP. PSO finds a feasible layout in only 19/30 runs from random starts (30/30 with feasible starts). The optimized layouts may use the benchmark spacing $4D$, whereas the installed grid keeps $7D$ (no run with $7D$ was made); the block is evaluated without the wakes of the other 64 turbines; and cables, foundations, loads and navigation, which also shaped the installed grid, are not modeled. The comparison therefore tests the optimizers; it does not show that the installed layout could be improved.

C. IEA37 Case Study 1

In IEA37 Case Study 1 the participants chose their own optimizers and budgets [20]. None of our methods reaches the best feasible published layouts (Table VIII; layouts: Section S-IV-C of the supplementary material). At 30,030 evaluations, the relative AEP difference to them is -1.33% (16 turbines) and -4.20% (36 turbines) for the best PSO-VNS layout and -3.90% and -6.51% for the mean over the 30 runs; the best PSO-VNS layout would rank third among 9 and seventh among 8 feasible submissions. The gap is larger for 36 turbines, where the best of our layouts comes from MS-SLSQP.

TABLE IX

ROBUSTNESS TO THE BENCHMARK MODEL: ALL FEASIBLE FINAL LAYOUTS RE-EVALUATED (NOT RE-OPTIMIZED). Δ : MEAN RELATIVE CHANGE OF THE OBJECTIVE (%); RANK: AVERAGE RANK OF PSO-VNS OVER THE 68 CASES; BEST: BEST-RANKED METHOD; $\bar{\tau}$: MEAN KENDALL CORRELATION BETWEEN THE BENCHMARK AND ALTERNATIVE ORDERINGS WITHIN A CASE; SAME: CASES (%) WITH UNCHANGED BEST METHOD. THE RANKS OF ALL METHODS ARE GIVEN IN TABLE S20

Model	Δ (%)		Rank	Best	$\bar{\tau}$	Same
	DS I	DS II				
Benchmark	+0.0	+0.0	1.82	PSO-VNS	–	–
Cubic curve	-17.7	-36.9	1.82	PSO-VNS	0.95	94
Cubic, 25 m/s cut-out	-21.5	-37.0	1.80	PSO-VNS	0.95	93
Gaussian wake	-0.3	-0.4	1.97	PSO	0.66	54

As the participants' budgets are not reported comparably, the size of the gap is indicative only. The 64-turbine scenario was not run, and, like the published layouts, ours are optimized for 16 discrete directions and one wind speed.

D. Computational Cost and Published Results

Per evaluation, the metaheuristics need 0.23–0.45 ms and MS-SLSQP, with its larger optimizer overhead, 0.53 ms, about 2.3 times as long as PSO (Table S19; indicative, measured on the machines that produced the runs); one 6,030-evaluation run of PSO-VNS takes 1.5–3.2 s, depending on N . Published Kusiak–Song results [5]–[8] are discussed in Section S-VII of the supplementary material; they are not compared numerically because budgets, constraint handling and reporting differ.

IX. SENSITIVITY TO THE MODELING ASSUMPTIONS

Power curve and wake model. To test whether our conclusions depend on the benchmark models, we re-evaluated every feasible final layout of the main comparison, without re-optimization, with 1) a cubic power curve [46], $f_c(s) = P_{\text{rated}}(s^3 - s_{\text{cut-in}}^3)/(s_{\text{rated}}^3 - s_{\text{cut-in}}^3)$ between the cut-in and rated speeds, 2) the same curve with a 25 m/s cut-out, and 3) the Gaussian wake model of [47] with growth rate $K_G = 0.04$ and the superposition and Weibull scaling of the benchmark (Section S-V-A of the supplementary material). The absolute energy depends strongly on the model (Table IX): the cubic curve gives 17.7% (Data Set I) and 36.9% (Data Set II) less expected power than the linear benchmark curve, so our energy values are benchmark values, not AEP predictions for a specific turbine. The ranking of the methods is stable under the cubic curves ($\bar{\tau} = 0.95$; same best method in 94% of the cases), and PSO-VNS keeps the best average rank. It is less stable under the Gaussian model ($\bar{\tau} = 0.66$; same best method in 54% of the cases): this model differs from the Jensen cone both in its smooth lateral profile and in its (uncalibrated) expansion rate, so layouts tuned to the Jensen cone are judged differently. PSO ranks first and PSO-VNS second, but the margin in average rank is small and untested (1.91 vs. 1.97), and both are ahead of all other methods, so the small Jensen-model advantage of the VNS phase over PSO does not carry over.

Spacing and boundary handling. The optimized layouts often lie on the 4D constraint inherited from the benchmark: of the feasible final layouts of the eight methods, 49% also satisfy 5D and 36% satisfy 6D (4.1% and 0.1% for $N = 11$ –15), so they cannot be transferred to sites with a larger spacing without re-optimization. At 5D and 6D, a multi-start packing search fits fewer turbines into each circle, and the largest 500-m cases may be infeasible (bounds and a re-optimization of six cases: Tables S21 and S22). The boundary rule also matters: an otherwise identical SSA that projects every turbine radially onto the circle instead of clipping the coordinates attains higher mean benchmark objectives than the original SSA in all six representative cases at 3,030 evaluations (Mann–Whitney test, $p < 0.01$ in every case). Because the boundary rule alone changes the results, all methods use the same rule (Section II-C).

X. DISCUSSION AND RECOMMENDATIONS

The three main findings (Sections VI–IX) share one implication: in the continuous WFLOP, which method appears best can depend on the comparison protocol as strongly as on the methods. Replacing the old setting, which violates the order-2 stability condition (8), by the constriction setting moves PSO from position 5 to 1 in Table I, ahead of every salp-swarm method, at 6,030 evaluations; at 120,030 evaluations, SSA-VNS ranks ahead of PSO on the six largest cases (4.17 vs. 5.83). An advantage over PSO is therefore informative only if the PSO baseline is convergent, and a swarm phase must beat a random-sampling control of equal cost. RS-VNS is a weak, no-search control (a uniform sample in the bounding square has all N turbines inside the circle only with probability $(\pi/4)^N$), so beating it is a minimum bar: the SSA phase clears it narrowly (case-mean difference -0.068 percentage points, $p = 4.5 \times 10^{-4}$; run level 2/66/0), the LX-SSA phase not detectably ($p = 0.72$), and the PSO phase clearly (-0.399 percentage points).

For benchmark-type problems of up to 36 turbines with random starts, PSO-VNS or a constriction PSO is a sound first choice ($\omega = 0.75$ beat the pre-specified 0.5 on twelve test cases; confirm it on the target problem), and from a feasible start VNS alone is competitive; for engineering design, the choice should be re-checked on a site-specific objective. For WFLOP comparisons, we recommend these established practices [22], all applied in this paper:

- 1) use baseline settings with documented convergence or standard defaults (for PSO, (8) or (7)) and report them;
- 2) give all methods the same number of evaluations, including the initial population and all gradient calls;
- 3) pair the runs by seed and initial population;
- 4) use feasibility-aware ranking and report the share of feasible runs;
- 5) test the case means and the runs of each case with a multiplicity correction; do not read non-significance as equivalence;
- 6) in ablations of hybrids, include a random-sampling control of equal cost and the local method alone;
- 7) report feasible-start and model-robustness checks; publish code and data.

XI. LIMITATIONS

The benchmark uses the top-hat Jensen wake, the linear Kusiak–Song power curve, coarse direction bins that optimized layouts can exploit, circular farms, a $4D$ spacing and a prescribed N . The 68 cases are not independent (six farm–wind combinations with nested N), so the p -values describe this benchmark, not a population of farms; the $N \geq 10$ split is post hoc. The 68-case results refer to 6,030 evaluations, the budget study covers only six cases and Horns Rev 1, and the old PSO setting was run only at 6,030 evaluations and only on our benchmark. Only the PSO coefficients were revisited; DE, VNS and $\omega = 0.5$ are untuned defaults. The pool lacks state-of-the-art optimizers such as CMA-ES [48] or L-SHADE, and MS-SLSQP uses finite-difference gradients on a discontinuous objective, so its rank says little about gradient-based methods. RS-VNS samples in the bounding square; a stronger control (sampling inside the site, or a feasibility repair) may narrow the gap to the PSO phase, and the component analysis does not separate faster feasibility from better exploration. From feasible starts, VNS ranks first and PSO and DE stall (Section VIII-A), so our statements on PSO-VNS refer to random starts. The Horns Rev 1 comparison tests the optimizers, not the installed design (Section VIII-B); on IEA37, the best PSO-VNS layouts at 30,030 evaluations reach 98.7% (16 turbines) and 95.8% (36 turbines) of the AEP of the best feasible published layouts. Wake-added turbulence and fatigue loads (critical at $4D$), wake steering, blockage, terrain, cables, grid connection, electrical losses and cost objectives are outside our scope; all AEP values are gross.

XII. CONCLUSION

Under a controlled protocol, three findings emerge. First, the baseline configuration changes the conclusions: the old PSO setting violates the order-2 stability condition and places PSO behind salp-swarm hybrids, whereas the constriction setting places it ahead of every salp-swarm method at 6,030 evaluations. Second, a VNS phase significantly improves the salp-swarm methods but gives PSO only a small, non-significant gain; a PSO phase gives VNS a clearly better start than random sampling, while salp-swarm phases add little beyond random sampling and the Laplace step makes the hybrid worse. Third, PSO-VNS, which combines two standard components and adds one parameter, is a consistent reference method: it is feasible in every run and has the best average rank (1.82), also on the six largest cases at all three budgets (exploratory; first in 3 of 6 cases at 120,030 evaluations). Yet it is not significantly different from PSO over the 68 cases ($p = 0.30$; run level 14/50/4), its advantage is concentrated in Data Set II with $N \geq 10$ (lower loss in 10 of 10 cases; exploratory), VNS alone is best from feasible starts, PSO ranks first by a small margin under the Gaussian re-evaluation, and on IEA37 PSO-VNS stays below the best published layouts. Future work will study a stagnation-triggered switch between the phases, stronger baselines and random-sampling controls, larger farms, higher-fidelity wake models and economic objectives.

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[TBD: acknowledgment, or delete this section]

DATA AVAILABILITY

The source code of all methods and of the benchmark evaluator, the per-run results of all studies (including the runs with the old PSO setting) and the analysis scripts are openly available at [TBD: Zenodo DOI]. Each run record gives the method, case, seed, final objective (AEP for Horns Rev 1), feasibility, final turbine coordinates and convergence curve, and the budget and initialization where these vary. The file `analysis/README_reproduce.md` lists the script that produces each table, figure and number of this paper and of its supplementary material, and `analysis/requirements.txt` pins the software versions. The Horns Rev 1 and IEA37 input data are taken from [44], [45]. The supplementary material gives the model details, the pseudocode of SSA and LX-SSA, the wind data, the per-case results, the coordinates of the best layouts and further figures.

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